MQTT Data Structures

Data contained in MQTT messages is formatted with JSON syntax.

Topic ID generation

Each MQTT topic includes a device-specific ID, which is generated from the 7-byte serial number of the device via a scrambling function.
The scrambling function consists of the following steps:

- Invert the order of all bits of each byte in the serial number
- Apply a XOR operation to each byte (with a XOR operand different for each byte)
- Scramble the order of the 7 bytes
- Convert each byte into a 2-character ASCII string
- Concatenate the strings from the 7 bytes, obtaining a 14-character string

The following code implements the scrambling algorithm in C:

```
uint8_t bitOrderInvert(uint8_t byte)
{
	uint8_t invertedByte = 0;
	int i;

	for (i = 0; i < 4; i++) {
		invertedByte |= (byte & (1 << i)) << (7 - 2 * i);
	}
	for (i = 4; i < 8; i++) {
		invertedByte |= (byte & (1 << i)) >> (2 * i - 7);
	}
	return invertedByte;
}

void hexToString(uint8_t hex, char *str)
{
	int digit;

	digit = (hex & 0xF0) >> 4;
	str[0] = (digit < 10) ? ('0' + digit) : ('a' + digit - 10);
	digit = hex & 0x0F;
	str[1] = (digit < 10) ? ('0' + digit) : ('a' + digit - 10);
}

void idCreate(uint8_t serNo, char *id)
{
```

```
    uint8_t scrambledSerNo[7];

    scrambledSerNo[0] = bitOrderInvert(serNo[6]) ^ 0x6E;
    scrambledSerNo[1] = bitOrderInvert(serNo[1]) ^ 0x14;
    scrambledSerNo[2] = bitOrderInvert(serNo[3]) ^ 0xDE;
    scrambledSerNo[3] = bitOrderInvert(serNo[2]) ^ 0xE5;
    scrambledSerNo[4] = bitOrderInvert(serNo[5]) ^ 0xAF;
    scrambledSerNo[5] = bitOrderInvert(serNo[7]) ^ 0x30;
    scrambledSerNo[6] = bitOrderInvert(serNo[4]) ^ 0x04;
    hexToString(scrambledSerNo[0], &id[0]);
    hexToString(scrambledSerNo[1], &id[2]);
    hexToString(scrambledSerNo[2], &id[4]);
    hexToString(scrambledSerNo[3], &id[6]);
    hexToString(scrambledSerNo[4], &id[8]);
    hexToString(scrambledSerNo[5], &id[10]);
    hexToString(scrambledSerNo[6], &id[12]);
    id[14] = '\0';
}
```

The 14-character string generated with this algorithm is referred to as <id> in the topic definitions in the rest of this document.

M5Stick-C Thermostat Controller (MINI)

Dynamic Data

- SmartPID is MQTT publisher
- Topic: smartpidM5/mini/<id>/dynamic
- MQTT message format in monitor mode:
  {
  "time": <timestamp>,
  "temp": <temp>, // only in heating/cooling/thermostatic mode
  "unit": <unit>, // only in heating/cooling/thermostatic mode
  "humidity": <rh>, // only in humidify/dehumidify/hygrostatic mode
  "mode": <mode>,
  "relay": <relay>,
  "runmode": "monitor"
  }
- MQTT message format in run mode:
  {
  "time": <timestamp>,
  "SP": <temp>, // only in heating/cooling/thermostatic mode
  "temp": <temp>, // only in heating/cooling/thermostatic mode
  "unit": <unit>, // only in heating/cooling/thermostatic mode
  "RH SP": <rh>, // only in humidify/dehumidify/hygrostatic mode
  "humidity": <rh>, // only in humidify/dehumidify/hygrostatic mode
  "mode": <mode>,
  "relay": <relay>,
  "runmode" :<run>
  }
- <timestamp> is an integer number of seconds
- <temp> is a floating-point number expressing a temperature value
- <unit> is a string reporting measurement unit {"C", "F"}
- <rh> is a floating-point number expressing a relative humidity value
- <mode> is a string whose possible values are {"heating", "cooling", "humidify", "dehumidify","off"}
- <relay> is a string whose possible values are {"on", "off"}
- <run> is a string reporting the run mode type ({"standard", "advanced"} when in heating/cooling/thermostatic mode, {"humidity"} when in humidify/dehumidify/hygrostatic mode)
- Example in monitor mode:
  {
  "time": 2345,

"unit": "C",
"temp": 34.9,
"relay": "on",
"runmode": "monitor"
}

- Example in run mode:
  {
  "time": 2345,
  "SP": 35,
  "temp": 34.6,
  "unit": "C",
  "mode": "heating",
  "relay": "on",
  "runmode": "standard"
  }

Events

Standard mode

- SmartPID is MQTT publisher
- Topic: smartpidM5/mini/<id>/events/standard
- MQTT message format:
  {
  "time": <timestamp>,
  "event": <event_type>
  }
- <timestamp> is an integer number expressing a number of seconds
- <event_type> is a string whose possible values are: "start", "stop", "pause", "resume", "power lost", "power restored", "CH1 SP reached", "socket connected", "socket disconnected", "BT05-xxxxxxxx low bat"
- Example:
  {
  "time": 2345,
  "event": "stop"
  }

Advanced mode

- SmartPID is MQTT publisher
- Topic: smartpidM5/mini/<id>/events/advanced
- MQTT message format:
  {

```
        "time": <timestamp>,
        "event": <event_type>
}
```

- <timestamp> is an integer number expressing a number of seconds
- <event_type> is a string whose possible values are: "profile <index>" (where <index> is the number of the profile selected by the user), "start", "stop", "pause", "resume", "power lost", "power restored", "soak <x>" (where <x> is an index ranging from 1 to 8), "ramp <y>" (where <y> is an index ranging from 1 to 7), "socket connected", "socket disconnected", "BT05-xxxxxxxx low bat"
- Example:

```
{
        "time": 2345,
        "event": "ramp 1"
}
```

Commands

- SmartPID is MQTT subscriber
- Topic: smartpidM5/mini/<id>/commands
- MQTT message format:

```
{
        "start": <run_mode>,
        "CH1 profile": <profile_num>, // only in heating/cooling/thermostatic mode
        "CH1 SP": <temp>,     // only in heating/cooling/thermostatic mode
        "RH SP": <rh>, // only in humidify/dehumidify/hygrostatic mode
        "stop": <stop>,
        "pause": <pause>,
        "status": <status>
}
```

- <run_mode> is a string whose valid values are "standard" and "advanced" (only in heating/cooling/thermostatic mode)
- <profile_num> is an integer number ranging from 1 to 10
- <temp> is a floating-point number expressing a temperature value; the temperature range is [-40, 120] if temperature unit is °C, [-40, 248] if temperature unit is °F
- <rh> is a floating-point number expressing a relative humidity value
- <stop> is a boolean value (true to stop the thermostat process, false is ignored)
- <pause> is a boolean value (true to pause the thermostat process, false to resume it)
- <status> is a boolean value (true to trigger message publishing to the status topic, false is ignored)
- All the above JSON attributes are optional, a given MQTT message may contain one or more attributes. One exception is when the "start" attribute is present: if its value is "advanced", then the "CH1 profile" attribute must be present

- Example:

```
{
        "pause": true
}
```

Status

- SmartPID is MQTT publisher
- Topic: smartpidM5/mini/<id>/status
- MQTT message format:

```
{
        "serial": <serial_number>,
        "SSID": <ssid>,
        "client": <ip_addr>,
        "AP": <socket_ip_addr>,
        "FW": <socket_fw>,
        "socket status": <status>
}
```

- <serial_number> is a string with format "aabbccddeeffgg"
- <ssid> is a string containing the name of the Wi-Fi network to which the controller is connected
- <ip_addr> is a string containing the IP address of the controller
- <socket_ip_addr> is a string containing the IP address of the socket paired with the controller
- <socket_fw> is a string containing the version of the socket firmware
- <status> is a string whose allowed values are “connected” and “disconnected”

Profiles

- SmartPID is MQTT subscriber on the topic: smartpidM5/mini/<id>/profiles/update/#
- SmartPID is MQTT publisher on the topic: smartpidM5/mini/<id>/profiles/<X> where <X> is the profile number (1-10)
- MQTT message format:

```
{
        "SP1": <temp>,
        "soak1": <duration>,
        "ramp1": <duration>,
        "SP2": <temp>,
        "soak2": <duration>,
        "ramp2": <duration>,
        "SP3": <temp>,
        "soak3": <duration>,
        "ramp3": <duration>,
```

```
        "SP4": <temp>,
        "soak4": <duration>,
        "ramp4": <duration>,
        "SP5": <temp>,
        "soak5": <duration>,
        "ramp5": <duration>,
        "SP6": <temp>,
        "soak6": <duration>,
        "ramp6": <duration>,
        "SP7": <temp>,
        "soak7": <duration>,
        "ramp7": <duration>,
        "SP8": <temp>
}
```

- <temp> is a floating-point number expressing a temperature value; its range is [-40, 120] if the temperature unit is °C, [-40, 248] if the temperature unit is °F
- <duration> is an integer number of seconds comprised between 0 and (30 * 24 * 60 * 60)
- All attributes are optional: if a given attribute is missing in a message sent to SmartPID, the profile is saved with a default value for that attribute; the default value for temperature is 40 °C, the default value for duration is 0 seconds
- Example:

```
{
        "SP1": 40,
        "soak1": 480,
        "ramp1": 240,
        "SP2": 50,
        "soak2": 420,
        "ramp2": 120,
        "SP3": 60,
        "soak3": 180,
        "ramp3": 540,
        "SP4": 70,
        "soak4": 780,
        "ramp4": 840,
        "SP5": 85
}
```

M5Stack Thermostat Controller (PRO)

Dynamic Data

- SmartPID is MQTT publisher
- Topics:
  - smartpidM5/pro/<id>/dynamic/CH1
  - smartpidM5/pro/<id>/dynamic/CH2
- MQTT message format in monitor mode:

```
{
        "time": <timestamp>,
        "temp": <temp>,
        "unit": <unit>,
        "runmode": "monitor"
}
```

- MQTT message format in run mode:

```
{
        "time": <timestamp>,
        "countdown": <timer_down>,
        "countup": <timer_up>,
        "SP": <temp>,
        "temp": <temp>,
        "unit": <unit>,
        "mode": <mode>,
        "pwm": <pwm>,
        “maxpwm”:<maxpwm>,
        "runmode" :<run>
}
```

- <timestamp> is an integer number of seconds
- <timer_down> is an integer number expressing the number of seconds remaining before the end of the process
- <timer_up> is an integer number expressing the number of seconds elapsed from the start of the process
- <temp> is a floating-point number expressing a temperature value
- <unit> is a string reporting measurement unit {"C", "F"}
- <mode> is a string whose possible values are {"heating", "cooling", "off"}
- <pwm> is an integer number expressing a percentage value 0 - 100
- <maxpwm> is an integer number expressing a percentage value 0 - 100
- <run> is a string reporting the run mode type {"standard", "advanced"}
- Example in monitor mode:

```
{
```

"time": 2345,
"unit": "C",
"temp": 34.9,
"runmode": "monitor"
}

- Example in run mode:
{
"time": 2345,
"countdown": 360,
"countup": 600,
"SP": 35,
"temp": 34.6,
"unit": "C",
"mode": "heating",
"pwm": 25,
"runmode": "standard"
}

Events

Standard mode

- SmartPID is MQTT publisher
- Topic: smartpidM5/pro/<id>/events/standard
- MQTT message format:
{
"time": <timestamp>,
"event": <event_type>
}
- <timestamp> is an integer number expressing a number of seconds
- <event_type> is a string whose possible values are: "start", "stop", "pause", "resume", "power restored", "CH1 SP reached", "CH2 SP reached", "CH1 timer expired", "CH2 timer expired", "CH1 timer reset", "CH2 timer reset", "BT05-xxxxxxxx low bat"
- Example:
{
"time": 2345,
"event": "stop"
}

Advanced mode

- SmartPID is MQTT publisher
- Topic: smartpidM5/pro/<id>/events/advanced

- MQTT message format:
  {
      "time": <timestamp>,
      "event": <event_type>
  }
- <timestamp> is an integer number expressing a number of seconds
- <event_type> is a string whose possible values are: "profile <index>" (where <index> is the number of the profile selected by the user), "start", "stop", "pause", "resume", "power restored", "soak <x>" (where <x> is an index ranging from 1 to 8), "ramp <y>" (where <y> is an index ranging from 1 to 7), "BT05-xxxxxxxx low bat"
- Example:
  {
      "time": 2345,
      "event": "ramp 1"
  }

Commands

- SmartPID is MQTT subscriber
- Topic: smartpidM5/pro/<id>/commands
- MQTT message format:
  {
      "start": <run_mode>,
      "CH1 profile": <profile_num>,
      "CH2 profile": <profile_num>,
      "CH1 SP": <temp>,
      "CH2 SP": <temp>,
      “CH1 maxpwm” : <maxpwm>,
      “CH2 maxpwm” : <maxpwm>,
      "CH1 countdown": <time>,
      "CH2 countdown": <time>,
      "stop": <stop>,
      "pause": <pause>,
      "resume": <resume>,
      "status": <status>
  }
- <run_mode> is a string whose valid values are "standard", "advanced" and "monitor"
- <profile_num> is an integer number ranging from 1 to 10
- <temp> is a floating-point number expressing a temperature value; the temperature range is [-200, 450] if temperature unit is °C, [-328, 842] if temperature unit is °F
- <maxpwm> is an integer number expressing a percentage value 0 - 100
- <time> is an integer number of seconds comprised between 1 and (99 * 60 * 60 + 59 * 60 + 59)

- <stop> is a boolean value (true to stop the thermostat process, false is ignored)
- <pause> is a boolean value (true to pause the thermostat process, false to resume it)
- <resume> is a boolean value (true to resume an interrupted process, false to discard it)
- <status> is a boolean value (true to trigger message publishing to the status topic, false is ignored)
- All the above JSON attributes are optional, a given MQTT message may contain one or more attributes. One exception is when the "start" attribute is present: if its value is "advanced", then the "CH1 profile" attribute must be present
- Example:

```
{
        "pause": true
}
```

Status

- SmartPID is MQTT publisher
- Topic: smartpidM5/pro/<id>/status
- MQTT message format:

```
{
        "serial": <serial_number>,
        "SSID": <ssid>,
        "client": <ip_addr>
}
```

- <serial_number> is a string with format "aabbccddeeffgg"
- <ssid> is a string containing the name of the Wi-Fi network to which the controller is connected
- <ip_addr> is a string containing the IP address of the controller

Profiles

- SmartPID is MQTT subscriber on the topic: smartpidM5/pro/<id>/profiles/update/#
- SmartPID is MQTT publisher on the topic: smartpidM5/pro/<id>/profiles/<X>
  where <X> is the profile number (1-10)
- MQTT message format:

```
{
        "SP1": <temp>,
        "soak1": <duration>,
        "ramp1": <duration>,
        "SP2": <temp>,
        "soak2": <duration>,
        "ramp2": <duration>,
        "SP3": <temp>,
        "soak3": <duration>,
```

"ramp3": <duration>,
"SP4": <temp>,
"soak4": <duration>,
"ramp4": <duration>,
"SP5": <temp>,
"soak5": <duration>,
"ramp5": <duration>,
"SP6": <temp>,
"soak6": <duration>,
"ramp6": <duration>,
"SP7": <temp>,
"soak7": <duration>,
"ramp7": <duration>,
"SP8": <temp>
}

- <temp> is a floating-point number expressing a temperature value; its range is [-200, 450] if the temperature unit is °C, [-328, 842] if the temperature unit is °F
- <duration> is an integer number of seconds comprised between 0 and (30 * 24 * 60 * 60)
- All attributes are optional: if a given attribute is missing in a message sent to SmartPID, the profile is saved with a default value for that attribute; the default value for temperature is 40 °C, the default value for duration is 0 seconds
- Example:

{
"SP1": 40,
"soak1": 480,
"ramp1": 240,
"SP2": 50,
"soak2": 420,
"ramp2": 120,
"SP3": 60,
"soak3": 180,
"ramp3": 540,
"SP4": 70,
"soak4": 780,
"ramp4": 840,
"SP5": 85
}